

Quantum Simulation of Extended Higgs-Sector Spectra for Dark-Energy Investigation

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Project Description

This project extends the computational framework being developed in *Full Spectrum Nonperturbative Higgs Vacuum Decay via Euclidean Lattice Simulation in 4D*. That work establishes a lattice–Euclidean workflow for constructing $O(4)$ -symmetric bounce solutions, discretizing the Euclidean action and fluctuation Hessian, isolating negative and translational zero modes, extracting the lowest positive eigenmodes, and computing determinant ratios. The framework also incorporates an RG-improved Higgs potential and provides a route to multi-field, finite-temperature, and curved-background calculations.

These prior results will supply validated sparse fluctuation operators and classical benchmark spectra that can be translated into quantum-simulatable Hamiltonians. The proposed work will implement block encoding, qubitization, Quantum Phase Estimation, and Variational Quantum Eigensolvers to study the low-energy spectra of extended Higgs sectors. Classical high-performance calculations will provide reference eigenvalues, error estimates, and training data for hybrid classical–quantum algorithms.

The central scientific objective is to determine whether Higgs-portal fields, multi-scalar interactions, or higher-dimensional operators can generate robust, parametrically light scalar modes. A low eigenvector of the fluctuation Hessian defines a collective field direction, while its eigenvalue measures the local curvature of the effective potential along that direction. Quantum spectral methods therefore provide a direct way to identify candidate light degrees of freedom that may be hidden within an extended Higgs sector.

The dark-energy connection will be evaluated within the quintessence framework, in which a slowly evolving scalar field Q has energy density $\rho_Q = \dot{Q}^2/2 + V(Q)$ and pressure $p_Q = \dot{Q}^2/2 - V(Q)$. When potential energy dominates, the equation-of-state parameter $w_Q = p_Q/\rho_Q$ approaches -1 , allowing the field to drive accelerated expansion. For each candidate eigenmode, the project will reconstruct the effective potential beyond its quadratic curvature, determine its effective mass and field excursion, and test slow-roll, equation-of-state, and energy-density requirements.

Candidate modes will also be tested for stability under changes in lattice spacing, volume, renormalization inputs, model parameters, and quantum-algorithm implementation. This is essential because an ultra-light eigenvalue alone does not establish a viable quintessence field: the mode must remain sufficiently flat, weakly coupled, radiatively stable, and compatible with cosmological evolution. The project will deliver quantum encodings, classical and quantum spectral benchmarks, parameter-space scans, and quantitative viability tests. A positive result would identify concrete Higgs-sector candidates for further dark-energy study; a null result would constrain this class of dynamical scalar-field explanations.

References

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Scientific Objectives

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1. Quantum Encoding of Higgs-Sector Operators

The previously developed fluctuation Hessians $H = \delta^2 S / \delta \phi^2$ will be translated into sparse quantum Hamiltonians using:

- Block-encoding techniques,
- Qubitization methods,
- Sparse-matrix Hamiltonian simulation frameworks.

These encodings will allow quantum algorithms to access the low-lying spectrum of extended Higgs-sector Hamiltonians. Each operator will be shifted and normalized to obtain a bounded Hermitian representation while preserving the ordering of the physically relevant low modes. Alternative encodings will be compared using qubit count, circuit depth, oracle complexity, and sensitivity to truncation error. The result will be a documented encoding workflow that can be reused across the model families considered below.

2. Quantum Low-Mode Extraction

Quantum Phase Estimation (QPE) and Variational Quantum Eigensolvers (VQE) will be used to extract the lowest 50–200 eigenvalues of the encoded Hamiltonians. These results will be benchmarked against the previously developed classical spectra to validate correctness.

Classically determined low-mode profiles will guide trial-state preparation and provide controlled test cases. QPE will target high-precision spectral resolution, while VQE will assess near-term implementations with shallower circuits. Convergence, degeneracies, near-zero modes, and accumulated errors from finite sampling, hardware noise, and imperfect state preparation will be quantified separately.

3. Extended Higgs-Sector Quantum Simulation

Hamiltonians will be constructed for:

- Higgs-portal scalar fields,
- Two-Higgs-doublet-like models,
- Higher-dimensional operators,
- Finite-temperature and curved-background corrections.

Parameter-space scans will be performed to identify emergent ultra-light modes.

The scans will vary portal masses and couplings, scalar self-interactions, higher-dimensional operator coefficients, temperature, and background curvature. A candidate mode will be retained only if it persists under changes in lattice resolution, finite volume, spectral truncation, and quantum encoding, thereby distinguishing physical light modes from numerical or algorithmic artifacts.

4. Dark-Energy Mode Identification

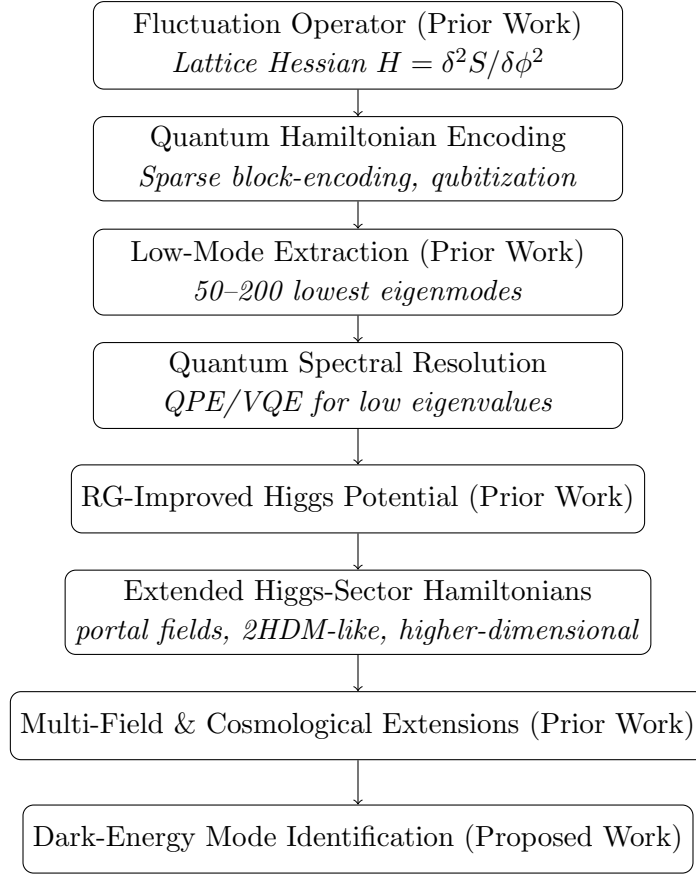
For any robust ultra-light eigenmode:

- Effective mass m_ϕ will be extracted,
- Potential curvature along the eigenvector direction will be analyzed,
- Slow-roll conditions will be evaluated,
- Equation-of-state evolution $w_Q(z)$ and energy density will be estimated,
- Radiative stability and phenomenological constraints will be assessed.

The normalized eigenvector will define a collective scalar coordinate whose reconstructed potential and homogeneous evolution will be tested against slow-roll, present-day energy-density, and accelerated-expansion requirements before the mode is identified as a quintessence candidate.

Project Summary Diagram

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Computational Resources

Classical HPC

Multi-node Intel/AMD CPU clusters and NVIDIA A100/H100 GPU nodes will support sparse eigensolvers (ARPACK, SLEPc, and PRIMME) and parallel linear algebra (PETSc and Trilinos). These calculations will provide benchmark spectra and training data for hybrid algorithms.

Primary Quantum Platform

IBM Quantum and Qiskit will support Hamiltonian simulation, QPE/VQE, error mitigation, and execution on superconducting-qubit hardware.

Cross-Validation

IonQ, Quantinuum, and Rigetti platforms will be used, when available, to compare hardware architectures and test the robustness of spectral results.